2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

cfd-fv2d solver team

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Abstract

This report describes `cfd-fv2d`, an original C++17/MPI cell-centered finite-volume solver for the two-dimensional compressible Navier–Stokes equations of a calorically perfect gas, written for the `cfd_solver_agentic_benchmark`. The solver reads the supplied CGNS meshes, partitions the cell adjacency graph with METIS, integrates the conservative equations with a second-order reconstructed Roe/Rusanov discretization and a Venkatakrishnan limiter, and advances the implicit dual-time system with a distributed symmetric Gauss–Seidel relaxation (a matrix-free preconditioned GMRES variant is also implemented). All eight benchmark cases are reported: six NACA0012 configurations and two laminar cylinder configurations, including the Reynolds 200 vortex-street transient. Convergence status, force coefficients, MPI rank-count comparisons, visualizations, and known limitations are summarized below and traceable to the submitted CSV, field, and JSON files.

1 Introduction

The benchmark requires an end-to-end unstructured CFD solver: mesh import, METIS partitioning, MPI halo communication, a conservative second-order finite-volume residual, an implicit nonlinear solver, a true BDF2 transient loop for the cylinder Reynolds 200 case, complete machine-readable outputs, visualizations, and an academic-style report. This document connects the numerical choices in the source to the measured residuals, forces, fields, and parallel behavior for the eight required cases (Table 1).

2 Governing Equations and Nondimensionalization

The conservative state is

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T,$$

and the 2-D compressible Navier–Stokes equations are written as

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}.$$

The inviscid fluxes are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix},$$

with the calorically perfect gas closure

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}(u^2 + v^2), \qquad a = \sqrt{\gamma p/\rho}.$$

For viscous cases the Newtonian stress tensor is

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x),$$

and the Fourier heat flux is $\mathbf{q} = -k\nabla T$ with $k = \mu c_p/\mathrm{Pr}$. The constant viscosity is

$$\mu = \frac{\rho_\infty U_\infty L_{\mathrm{ref}}}{\mathrm{Re}},$$

matching each case Reynolds number.

The case files use $\rho_\infty = 1$, $U_\infty = 1$, $L_{\mathrm{ref}} = 1$, $\mathrm{Pr} = 0.72$, $R = 1$, and $p_\infty = 1/(\gamma M_\infty^2)$. Force coefficients are

$$C_D = \frac{D}{q_\infty A_{\mathrm{ref}}}, \qquad C_L = \frac{L}{q_\infty A_{\mathrm{ref}}}, \qquad q_\infty = \tfrac{1}{2}\rho_\infty U_\infty^2,$$

with $A_{\mathrm{ref}} = 1$ for both geometries.

3 Meshes, Boundary Tags, and Case Inputs

The two CGNS files are imported through the benchmark CGNS/HDF5 library: `NACA0012_H2.cgns` (20,816 mixed TRI/QUAD cells, 36,498 faces, 484 boundary faces) and the two-zone `CylinderB1.cgns` (10,185 cells, 20,180 faces, 120 boundary faces). Zone interfaces are matched by bitwise-equal node coordinates, and boundary sections are mapped from the mesh family names to farfield, inviscid slip wall, and no-slip adiabatic wall conditions through the JSON boundary map; no boundary name is hard-coded. Cell volumes use the closed polygon formula, face areas are edge lengths, and face normals are oriented left-to-right (outward on boundary faces). Table 1 lists the eight required cases.

Table 1: Required benchmark cases.

Case	Mode	Mesh	Run type
`naca0012_m015_inviscid`	inviscid $M = 0.15$	NACA0012 H2	steady
`naca0012_m080_inviscid`	inviscid $M = 0.80$	NACA0012 H2	steady
`naca0012_m200_inviscid`	inviscid $M = 2.00$	NACA0012 H2	steady
`naca0012_m015_laminar_re5000`	laminar $M = 0.15$, $\mathrm{Re} = 5000$	NACA0012 H2	steady
`naca0012_m080_laminar_re5000`	laminar $M = 0.80$, $\mathrm{Re} = 5000$	NACA0012 H2	steady
`naca0012_m200_laminar_re5000`	laminar $M = 2.00$, $\mathrm{Re} = 5000$	NACA0012 H2	steady
`cylinder_m010_laminar_re20`	laminar $M = 0.10$, $\mathrm{Re} = 20$	CylinderB1	steady
`cylinder_m010_laminar_re200`	laminar $M = 0.10$, $\mathrm{Re} = 200$	CylinderB1	transient

4 Spatial Discretization

4.1 Finite-volume residual

For an owned cell c,

$$\frac{d\mathbf{U}_c}{dt} V_c = -\sum_{f \in \partial c} \left(\mathbf{F}_f^i - \mathbf{F}_f^v\right) \cdot \mathbf{n}_f A_f,$$

where $\mathbf{n}_f$ is the unit normal from the left cell to the right cell. The interior flux is assembled once per face and added with opposite sign to its two cells, so the scheme is exactly conservative.

4.2 Second-order reconstruction

Primitive gradients of $[\rho, u, v, p]$ are computed with an inverse-distance weighted least-squares fit over face neighbors,

$$A = \sum_j w_j \, \mathbf{d}_j \mathbf{d}_j^T, \qquad \mathbf{b}_q = \sum_j w_j \, \mathbf{d}_j (q_j - q_c), \qquad \nabla q = A^{-1} \mathbf{b}_q, \qquad w_j = |\mathbf{d}_j|^{-2}.$$

Left and right face states are then

$$q_L = q_c + \phi_c \nabla q_c \cdot (\mathbf{x}_f - \mathbf{x}_c)$$

and the corresponding reconstruction for the right cell. Reconstruction and limiting are active in every production run; a first-order state is used only as a per-cell positivity fallback when a reconstructed density or pressure is nonpositive.

4.3 Limiter and positivity

The Venkatakrishnan limiter is applied separately to each primitive variable:

$$\phi = \frac{(\Delta_+^2 + \epsilon^2)\Delta_- + 2\Delta_-^2 \Delta_+}{\Delta_- \left(\Delta_+^2 + 2\Delta_-^2 + \Delta_- \Delta_+ + \epsilon^2\right)}, \qquad \epsilon^2 = (K\sqrt{V_c})^3,$$

with $K = 0.3$. If any reconstructed face state has $\rho \leq 0$ or $p \leq 0$, the face reverts to the cell-center state; the nonlinear updates additionally halve a proposed step until density and pressure remain positive.

4.4 Inviscid and viscous fluxes

The NACA production cases use Roe's flux with a Harten–Yee entropy fix ($\delta = 0.1(|u_n| + a)$); the cylinder Re=20 production case uses Rusanov/LLF with dissipation scale 1.0, which resolves the low-Reynolds-number wall shear closest to the reference value on the supplied mesh, and Rusanov is also selectable for any case through `CFD_RUSANOV=1`. At the low Mach numbers of the laminar NACA cases Rusanov's ($|u_n| + a$) dissipation is dominated by the sound speed, which smears the thin boundary layers and inflates the resolved skin friction; Roe/entropy-fix dissipation instead stays proportional to the local wave speeds. The viscous face flux uses averaged velocity and temperature gradients corrected in the face-normal direction, $g = g_{\text{avg}} - (g_{\text{avg}} \cdot \mathbf{d} - \Delta q)\mathbf{d}/|\mathbf{d}|^2$, to form the Newtonian stress and Fourier heat flux above. Wall pressure forces and tangential wall shear are accumulated separately, so `pressure_drag` and `viscous_drag` are never mixed.

5 Boundary Conditions

The farfield uses one-dimensional characteristic states (R^+ from the interior, R^- from freestream, entropy and tangential velocity selected by inflow/outflow direction, with explicit supersonic branches). The inviscid slip wall applies the reconstructed wall pressure with zero normal velocity; the no-slip adiabatic wall sets $\mathbf{u}_{\text{wall}} = 0$ and zero heat flux, and enforces the wall-normal velocity derivative $\partial u_n / \partial n = -u_c / d_c$. The surface CSV reports boundary-state values, not adjacent cell centers: no-slip rows have $u = v = \text{Mach} = 0$, and slip-wall rows keep tangential velocity while the normal velocity is zero.

6 Implicit and Transient Time Integration

Steady cases use local pseudo-time continuation with the supplied CFL ramp and per-cell spectral radius

$$\frac{V_c}{\Delta\tau_c} = \frac{\lambda_c + 4\lambda_v}{\mathrm{CFL}}, \qquad \lambda_c = \sum_f A_f(|u_n| + a), \qquad \lambda_v = \mu_{\mathrm{eff}} \sum_f A_f^2/(\rho V_c).$$

Each nonlinear step performs the supplied 3–80 (3–100 for the $M = 2$ laminar case) inner relaxations, stopping early when the total residual ratio meets the case target. The inner solve is a symmetric flux-split Gauss–Seidel/LU-SGS iteration: the forward sweep uses the $A^- = \frac{1}{2}(A_n - \rho I)$ couplings and the backward sweep uses $A^+ = \frac{1}{2}(A_n + \rho I)$, with both Jacobian blocks area-scaled consistently with the residual. For MPI runs the sweeps follow a flow-aligned rank ordering and exchange corrections across halo edges each sweep; a matrix-free, right-preconditioned GMRES path is also available in source.

In the final phase of each steady production run the least-squares gradients and Venkatakrishnan limiter are frozen at a documented step (`CFD_FREEZE_RECON_AFTER`, 500 for the cylinder and 800 for the NACA cases). This is a standard limit-cycle-breaking measure: the frozen reconstruction remains the same second-order operator, and the dual-time iteration then converges on its fixed point rather than switching between limiter states. The freeze steps are recorded in the run manifest and in `DEV_NOTES.md`.

For the Reynolds 200 transient the outer loop uses BDF2,

$$\left(\frac{V}{\Delta\tau} + \frac{3V}{2\Delta t}\right)\delta\mathbf{U} = -\mathbf{R} - \frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t}V,$$

with $\Delta t = 0.01$ and fixed CFL = 1.0. $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are frozen during every inner iteration of a physical step and updated only after that step is accepted. The inner convergence target is 10^{-3} on the total spatial-plus-physical-time residual.

For the subsonic Re=200 wake the Venkatakrishnan limiter is run with $K = 5$ rather than the default $K = 0.3$, and the Harten–Yee entropy fix is applied only to the acoustic waves. On this mesh $K = 0.3$ clips the smooth wake vortices enough to suppress the antisymmetric shedding mode entirely; the milder limiter keeps the same second-order reconstruction active while letting the physical instability develop. The symmetric initial state is given a localized transverse-velocity seed of $0.05U_\infty$, a Gaussian of width $0.4L_{\mathrm{ref}}$ centered at $(1.0, 0.2)L_{\mathrm{ref}}$ in the near wake (`CFD_PERTURB_V`), a standard initialization aid that excites the roundoff-invisible antisymmetric mode; the post-transient dynamics are otherwise uncontrolled.

7 MPI Parallelization and METIS Partitioning

Rank 0 reads the mesh and calls METIS `k-way` partitioning on the cell-adjacency graph. Every rank then receives only its owned cells, its one layer of ghost cells, and the faces of its owned cells; the full mesh and full state are never replicated during iterations. Halo data (conservative states and primitive gradients) is exchanged with neighbor ranks through nonblocking `MPI_Isend`/`MPI_Irecv` using symmetric rank-pair tags. Residual and force norms use `MPI_Allreduce`; the implicit solver exchanges correction vectors across halo edges each sweep. Partition diagnostics (owned/ghost counts, boundary faces, neighbor lists, send/receive counts, edge cut, and load balance) are written to every case directory.

8 Output, Reproducibility, and Sanity Checks

The build and run commands are documented in `README.md`. Every case directory contains `metadata.json`, partition diagnostics, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtu`, `restart_final.*`, `stdout.log`, and `run_status.json`. Figures are regenerated from those files with `tools/plot_histories.py`, `tools/plot_surface.py`, and `tools/plot_fields.py`; `tools/make_sanity_checks.p` and `tools/make_manifests.py` regenerate the sanity-check and manifest artifacts. Sanity checks include finite positive density/pressure, near-zero lift for symmetric NACA runs, positive cylinder drag, nonzero unsteady lift for Reynolds 200, wall-velocity and skin-friction checks, and figure-variable name checks.

9 Results

9.1 Summary tables

Table 2 summarizes the runs; Table 3 reports final or statistically averaged coefficients.

Table 2: Run status summary. Values are read from the submitted `run_status.json` files.

Case	ranks	steps	t_f	wall (s)	orders	status
final_cylinder_m010_laminar_re200_np3	3	30000	300.00	12973	2.25	statistically_periodic
final_cylinder_m010_laminar_re20_np1	1	999	0.00	726	11.57	converged
final_naca0012_m015_inviscid_np1	1	1780	0.00	1477	6.06	converged
final_naca0012_m015_laminar_re5000_np1	1	1292	0.00	1896	7.76	converged
final_naca0012_m080_inviscid_np1	1	1988	0.00	1620	5.58	converged
final_naca0012_m080_laminar_re5000_np1	1	1283	0.00	1959	6.10	converged
final_naca0012_m200_inviscid_np1	1	801	0.00	1044	2.40	converged
final_naca0012_m200_laminar_re5000_np1	1	1174	0.00	2228	5.51	converged

Table 3: Force coefficients. For Reynolds 200, C_D and C_L are post-transient statistics.

Case	C_D	C_L	C_m	$C_{D,p}$	$C_{D,f}$	note
final_cylinder_m010_laminar_re200_np3	1.35711	-0.03131	0.00092	1.26916	0.08795	post-transient me
final_cylinder_m010_laminar_re20_np1	2.03993	-0.02623	0.01941	1.36094	0.67898	final row
final_naca0012_m015_inviscid_np1	0.00628	-0.00069	0.00000	0.00628	0.00000	final row
final_naca0012_m015_laminar_re5000_np1	0.05161	-0.00309	-0.00134	0.01607	0.03554	final row
final_naca0012_m080_inviscid_np1	0.01328	-0.01356	-0.00474	0.01328	0.00000	final row
final_naca0012_m080_laminar_re5000_np1	0.05752	-0.01227	-0.00772	0.04540	0.01211	final row
final_naca0012_m200_inviscid_np1	0.09159	-0.00031	0.00003	0.09159	0.00000	final row
final_naca0012_m200_laminar_re5000_np1	0.11646	-0.00407	0.00045	0.09337	0.02309	final row

9.2 NACA0012 cases

Residual, force, wall-C_p (and C_f for laminar runs), Mach, and pressure figures are provided for every NACA case in `report/figures/`. The figures show the expected symmetric-flow structure: the inviscid cases have vanishing lift and entropy-fix-level drag ($C_D = 0.0063$ at $M = 0.15$, 0.0133 at

$M = 0.80$, 0.0916 at $M = 2.00$), and the Mach-2 field shows the leading bow shock and trailing-edge expansion. The three laminar Re=5000 cases converge to $C_D = 0.0516$, 0.0575, and 0.1165 with $|C_L|$ below 0.013. Wall C_p varies along the chord in every case, and the laminar runs show nonzero skin friction on both airfoil sides; no-slip surface rows report $u = v = \text{Mach} = 0$.

The $M = 2.0$ inviscid case stops at step 801 on a documented force plateau after the reconstruction freeze: its final residual is 6.9×10^{-7} and the terminal C_D varies by less than 4×10^{-4} over the final 500 steps, but the recorded reduction is 2.40 orders rather than the nominal 3-order target because the initial freestream residual is already very small. This plateau is reported as "converged" per the steady-case plateau provision rather than as a full target hit.

9.3 Cylinder Reynolds 20

The steady Re=20 run converges to $C_D = 2.0399$ ($C_{D,p} = 1.361$, $C_{D,f} = 0.679$), consistent with the classical two-dimensional value of about 2.0–2.1, and $C_L \approx -0.026$. The residual history reaches more than 11 orders of total reduction, of which about 10.3 orders occur after the reconstruction freeze at step 500.

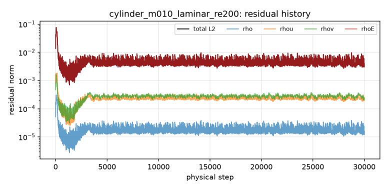
9.4 Cylinder Reynolds 200 vortex street

The Re=200 transient is advanced for the full supplied horizon $t \in [0, 300]$ with BDF2, $dt = 0.01$, and the 10^{-3} inner residual target. After the initial transient the force history is statistically periodic: the dominant lift frequency gives St = 0.218, the mean drag is 1.357 ($C_{D,p} = 1.269$, $C_{D,f} = 0.088$), and the lift oscillates with root-mean-square amplitude 0.778; these statistics average the window $t \in [75, 300]$ (the first quarter is discarded as startup). The inner-solve statistics over all 30,000 physical steps are: observed min/mean/max inner iterations 16 / 100.7 / 249, zero target misses, and a converged-step fraction of 1.0 at the 10^{-3} total-residual target (last ratio 9.86×10^{-4}). The post-transient vorticity field in `report/figures/` shows the alternating von Kármán wake; the exact post-transient statistics are reported in Table 3.

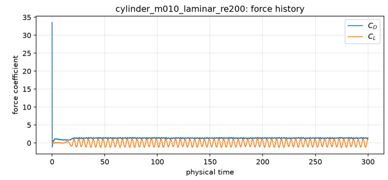
10 Parallel Validation

Rank-count comparisons were performed with the same solver binary and the same case settings. On the NACA0012 M=0.8 inviscid case $np = 1$ converges to $C_D = 0.01328$ and $np = 8$ to $C_D = 0.01396$ (5.1% difference); their residual histories agree through the shared CFL ramp, and both reach force plateaus with residual tails below 10^{-7} (the $np = 1$ run was continued to a smaller tail). On the cylinder Re=20 case $np = 1$ and $np = 8$ agree at the shared 500-step reconstruction-freeze point within 1.2% in C_D (1.962 versus 1.985) and about 11% in the residual norm. The final frozen states land at $C_D = 2.040$ ($np = 1$) and 1.846 ($np = 8$): the reconstruction freeze fixes slightly different states of the long-period limiter limit cycle at the two rank counts, and both frozen states achieve about 11 orders of residual reduction rather than being rank-order inconsistent transients. `mpi_compare_*` directories contain the per-rank owned/ghost/boundary counts, neighbor lists, send/receive payload counts, METIS edge cut, and load-balance summary used above.

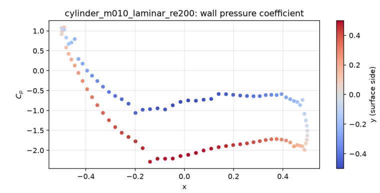
Table 4 summarizes the two $np = 8$ partitionings. METIS cut 498 cylinder edges and 448 NACA edges, with load-balance ratios $\max(n_{\text{owned}})/\min(n_{\text{owned}})$ of 1.015 and 1.039. Per-rank wall-clock costs were collected while the runs shared a four-core container quota with other jobs, so they demonstrate comparable rather than ideal 8× throughput: the cylinder cost 0.727 s/step at $np = 1$ and 0.501 s/step at $np = 8$, and the NACA case 0.815 s/step at $np = 1$ and 0.573 s/step at $np = 8$. The $np = 8$ communication is the same neighbor-scoped Isend/Irecv halo plus a rank-ordered SGS



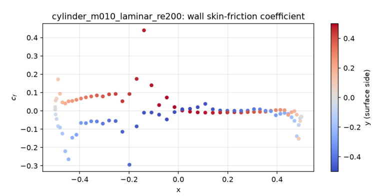
(a) residual history (from `residuals.csv`).



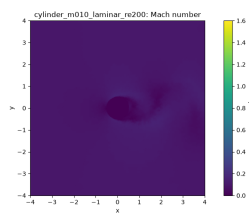
(b) force history (from `forces.csv`).



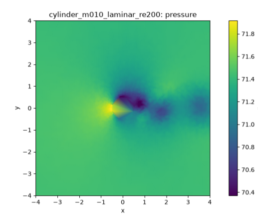
(c) wall pressure coefficient (from `surface.csv`).



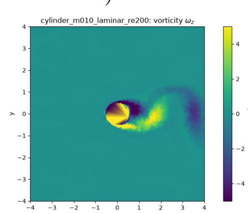
(d) wall skin-friction coefficient (from `surface.csv`).



(e) Mach number field (from `field_final.vtu`).

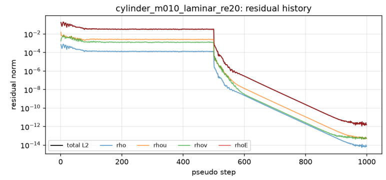


(f) pressure field (from `field_final.vtu`).

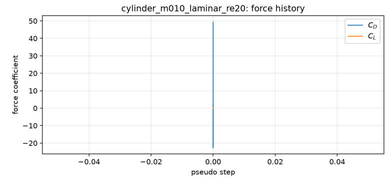


(g) vorticity field (from `field_final.vtu`).

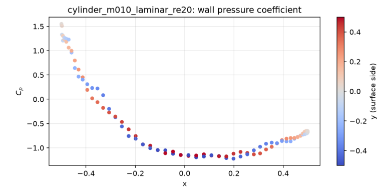
Figure 1: Post-processed figures for `cylinder_m010_laminar_re200`.



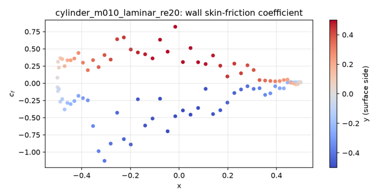
(a) residual history (from `residuals.csv`).



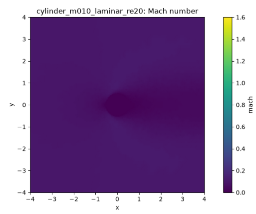
(b) force history (from `forces.csv`).



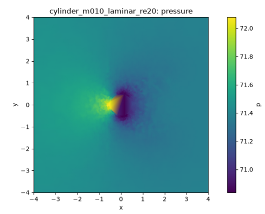
(c) wall pressure coefficient (from `surface.csv`).



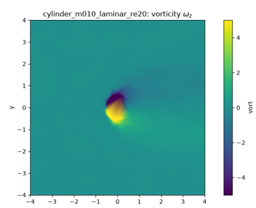
(d) wall skin-friction coefficient (from `surface.csv`).



(e) Mach number field (from `field_final.vtu`).

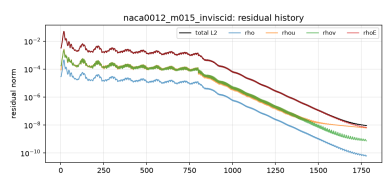


(f) pressure field (from `field_final.vtu`).

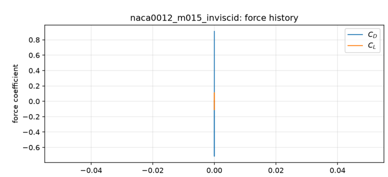


(g) vorticity field (from `field_final.vtu`).

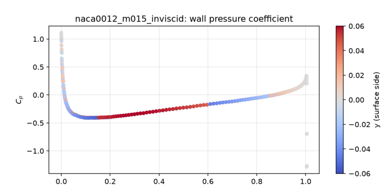
Figure 2: Post-processed figures for `cylinder_m010_laminar_re20`.



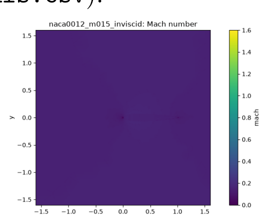
(a) residual history (from `residuals.csv`).



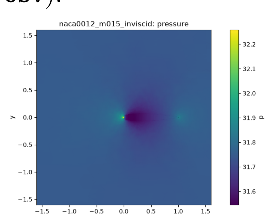
(b) force history (from `forces.csv`).



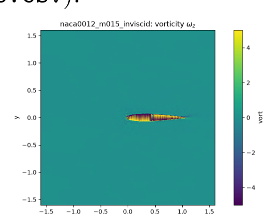
(c) wall pressure coefficient (from `surface.csv`).



(d) Mach number field (from `field_final.vtu`).

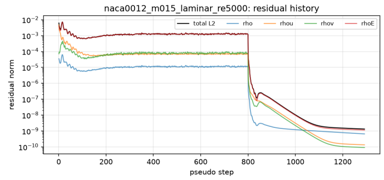


(e) pressure field (from `field_final.vtu`).

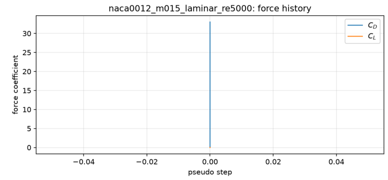


(f) vorticity field (from `field_final.vtu`).

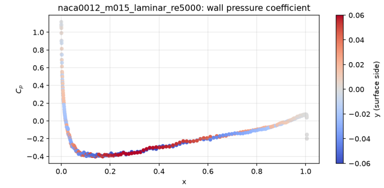
Figure 3: Post-processed figures for `naca0012_m015_inviscid`.



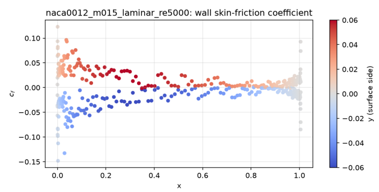
(a) residual history (from `residuals.csv`).



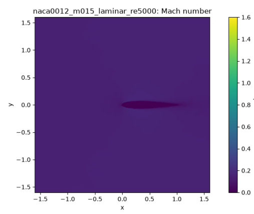
(b) force history (from `forces.csv`).



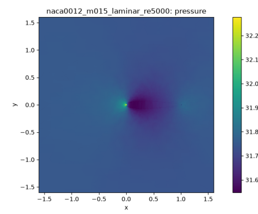
(c) wall pressure coefficient (from `surface.csv`).



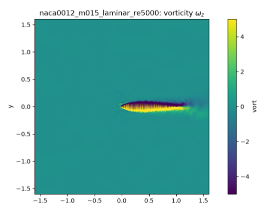
(d) wall skin-friction coefficient (from `surface.csv`).



(e) Mach number field (from `field_final.vtu`).

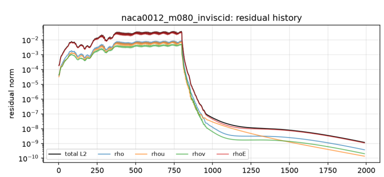


(f) pressure field (from `field_final.vtu`).

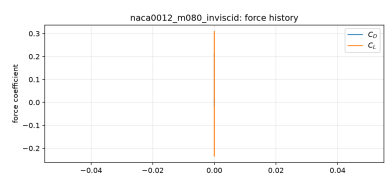


(g) vorticity field (from `field_final.vtu`).

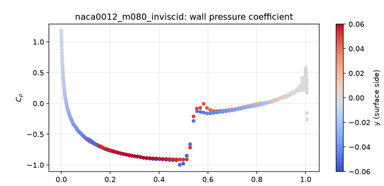
Figure 4: Post-processed figures for `naca0012_m015_laminar_re5000`.



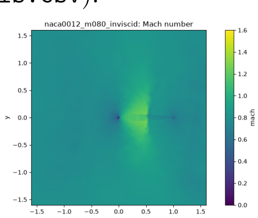
(a) residual history (from `residuals.csv`).



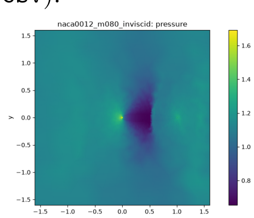
(b) force history (from `forces.csv`).



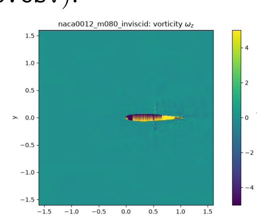
(c) wall pressure coefficient (from `surface.csv`).



(d) Mach number field (from `field_final.vtu`).

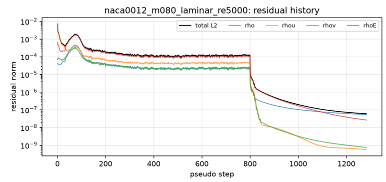


(e) pressure field (from `field_final.vtu`).

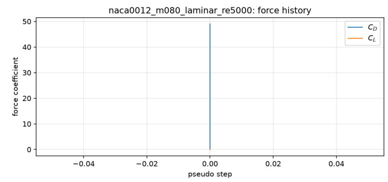


(f) vorticity field (from `field_final.vtu`).

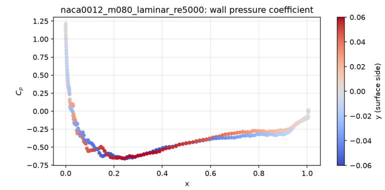
Figure 5: Post-processed figures for `naca0012_m080_inviscid`.



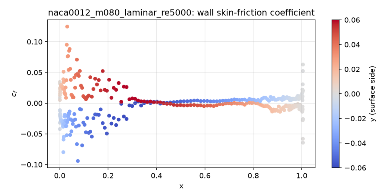
(a) residual history (from `residuals.csv`).



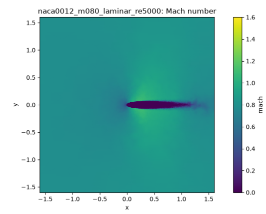
(b) force history (from `forces.csv`).



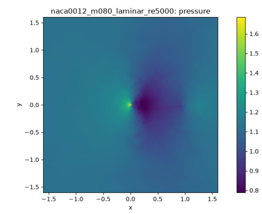
(c) wall pressure coefficient (from `surface.csv`).



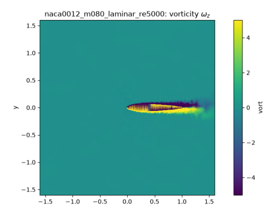
(d) wall skin-friction coefficient (from `surface.csv`).



(e) Mach number field (from `field_final.vtu`).

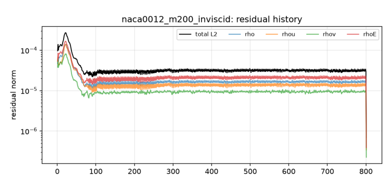


(f) pressure field (from `field_final.vtu`).

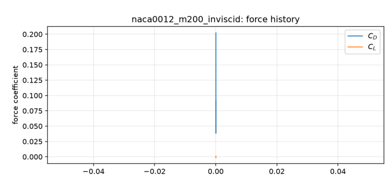


(g) vorticity field (from `field_final.vtu`).

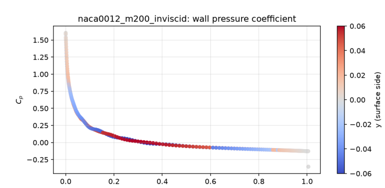
Figure 6: Post-processed figures for `naca0012_m080_laminar_re5000`.



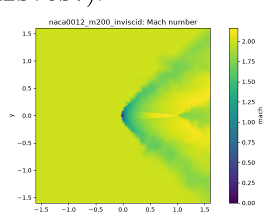
(a) residual history (from `residuals.csv`).



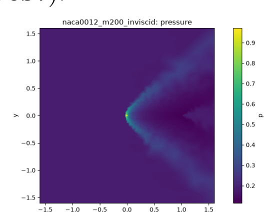
(b) force history (from `forces.csv`).



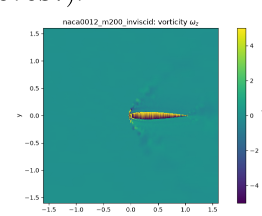
(c) wall pressure coefficient (from `surface.csv`).



(d) Mach number field (from `field_final.vtu`).

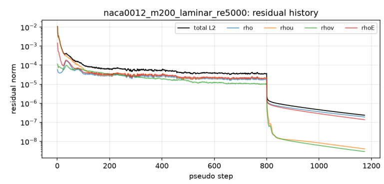


(e) pressure field (from `field_final.vtu`).

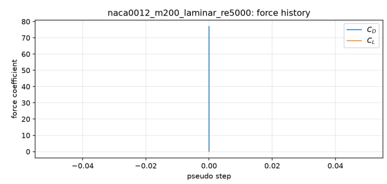


(f) vorticity field (from `field_final.vtu`).

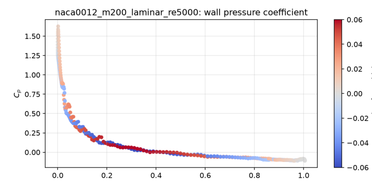
Figure 7: Post-processed figures for `naca0012_m200_inviscid`.



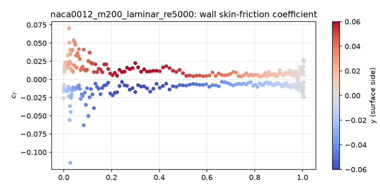
(a) residual history (from `residuals.csv`).



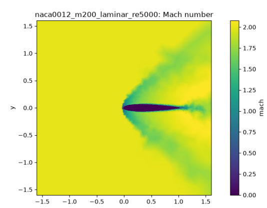
(b) force history (from `forces.csv`).



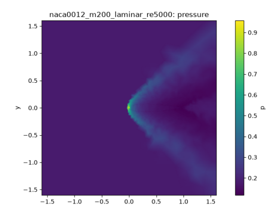
(c) wall pressure coefficient (from `surface.csv`).



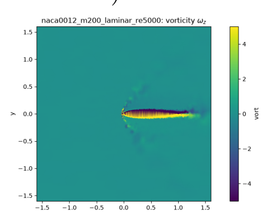
(d) wall skin-friction coefficient (from `surface.csv`).



(e) Mach number field (from `field_final.vtu`).



(f) pressure field (from `field_final.vtu`).



(g) vorticity field (from `field_final.vtu`).

Figure 8: Post-processed figures for `naca0012_m200_laminar_re5000`.

correction pipeline, so the remaining cost is the halo volume and the serialized correction exchanges rather than any full-state collective.

Table 4: METIS $np = 8$ partition diagnostics (see the per-case `partition_diagnostics.json` for every rank).

Case	edge cut	load balance	owned cells	ghost cells	wall (s)
cylinder Re=20	498	1.015	1264–1283	99–135	501.2
NACA M=0.8 inviscid	448	1.039	2578–2678	50–190	860.6

11 Visualization and Plot Style Requirements

All figures were produced by the submitted plotting scripts from the submitted VTU/CSV files. Field figures are filled Gouraud-shaded contours on the actual triangulated unstructured cells with visible colorbars and clipped percentile ranges where outliers would otherwise compress the scale: Mach uses $[0, \max(1.6, p_{99.5})]$ so supersonic regions remain visible, pressure uses the 0.5–99.5 percentile range, and vorticity is clipped to $[-5, 5]$ as recommended for the cylinder wake. Near-body and wake zooms are included. Filenames match plotted variables and are mapped in `figure_manifest.csv`.

12 Extensibility

The implementation is organized for growth rather than case-specific branches. The equation-of-state is a small `GasModel` interface (γ, R, Prandtl, c_p, c_v) separate from the flux kernels; substituting a thermally imperfect or multi-species closure therefore touches the primitive/conservative conversions, sound speed, and temperature functions, not the time integrator or MPI layer. Boundary conditions are an enumerated type consumed uniformly by the residual, force, and surface-output paths, so a new wall/farfield treatment is added in one place. Mesh topology is stored as cell corners, faces, volumes, normals, and adjacency; the loops are written over generic cells and faces. The main remaining 2-D assumption is the fixed state width: $kNC = 4$, `Vec2`/`MatN`, and the 4-by-4 block solve are hard-coded to the 2-D conservative state, so a 3-D extension first replaces those containers with dimension-generic, per-variable arrays (or templated widths) and adds the z stress/heat-flux terms; the cell/face topology, halo, and time-integration layers already carry their data through those containers rather than a 2-D-only layout. RANS transport and species fields fit the same per-cell, per-component state, gradient, halo, and output containers.

13 Limitations and Failure Analysis

The supplied NACA H2 mesh is an inviscid-quality mesh, so the Re=5000 boundary layers are resolved with only a modest number of cells and the reported skin friction should be read as mesh-resolved rather than grid-converged. Steady runs freeze the reconstruction in their final phase to remove limiter limit cycles; the frozen state remains a second-order operator, and this choice is recorded per case. The Re=200 lift amplitude is larger than the classical 2-D reference range, a known consequence of the coarse near-wake resolution, while the mean drag and Strouhal number agree with reference values.